Introduction to EXP2-1 Part 2

EXP2-1 Same or Different?

Semester B, Weeks 1-2

The **aim** of EXP2-1 is to develop tools and skills, in order to:

- Present and report data in the correct way

  (units, significant figures, error, and others)
- Determine whether the data are statistically significant

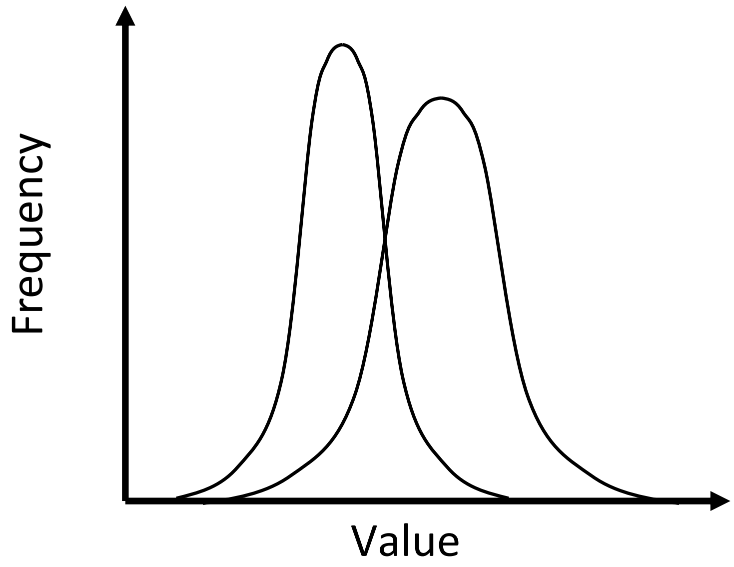
Introduction to EXP2-1 Part 2

Part 2 Significance tests

(comparing results, or samples)

Introduction to EXP2-1 Part 2

Significance tests



Are these 2 sets of data different from each other?

Significance tests

3 ways to determine differences between sets of data that vary randomly:

z-test - sample different from a population

t-test - samples different from each other

F-test - samples have different variance from each other

The z-test

Is a sample different from a known population?

You can knowledge about a population from a previous sample, if you had enough data.

e.g. Are your measured values different from known values?

Used to compare new values to **a <u>large</u> existing data set**

Used in process control to see if a process has changed

Some definitions

Null hypothesis (H_0) : a type of hypothesis stating that there is no difference between two samples derive from the same general population.

Significance level (α): the probability of rejecting the null hypothesis when it is true.

z standard score: a measure of how many standard deviations a sample mean is from the population mean.

α - significance level e.g. $\alpha = 0.05$, 5% level, i.e. $z = 1.96$

$\alpha = 0.01$, $z = 2.58$

$\alpha = 0.003$, $z = 3.00$

Formula – z standard score

Mean of sample

Mean of population of comparison

z - standard score

$$z = \frac{\overline{X} - \mu}{s_{\overline{X}}}$$

SD sample x

z > 0, sample mean > population mean;

z ~ 0, sample mean ~ population mean;

z < 0, sample mean < population mean.

Testing a null hypothesis

"The means of the sample and population are the same"
(Or rather: if the means are different, it is only out of chance,
this is the '**null hypothesis' (H_0)**)

Set a significance level, $\alpha = 0.05$, $z = 1.96$ (critical z)

i.e. 95 times out of 100 your hypothesis is correct

If more than 5% chance of being wrong, reject the null hypothesis

If $z < 1.96$, null hypothesis is correct, sample and population are the same

If $z > 1.96$, null hypothesis incorrect, it is "statistically significant"

("we can gain confidence") that sample and population are different

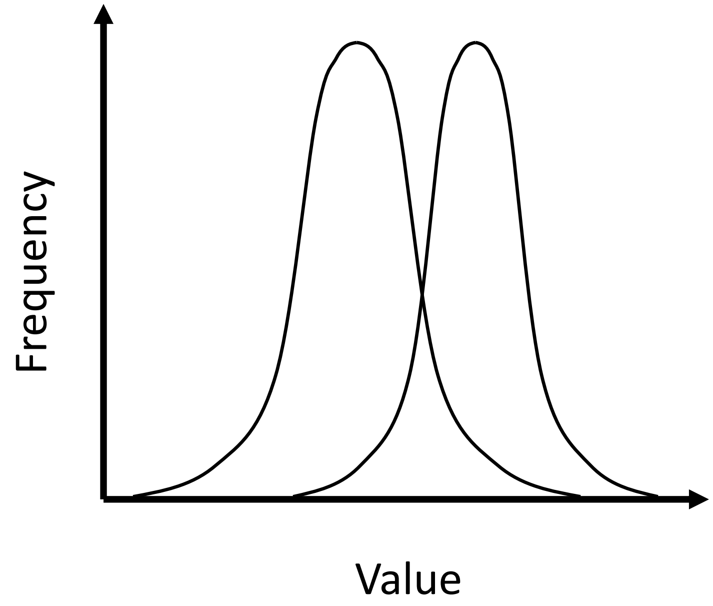
WE CAN ONLY TEST the null Hypothesis

Z-test conditions

- Compare whether there is significant difference between the mean values of two samples (normally for large samples, n>30);
- Compare whether there is significant difference between a new data and a population when the SD of the population is known.

Student’s t-test

Are two samples different from each other?



e.g. Are data from a control group and experimental group different?

Has a pharmaceutical drug had an effect?

Are men and women equally tall?

Tests

Comparing different types of variables

- One **numerical** variable (e.g. height)
- One **categorical** variable (groups A or B, women or men, etc.)
- Comparisons of means, medians and distributions
- **t tests**, Mann-Whitney, ANOVA

Two **categorical** variables:

- Cross-tabulation & proportions
- Chi-squared & McNemar tests

Two **numerical** variables:

- Scatter plots
- Regression & correlation

Student's t-test

Research questions:

- Does the **typical** value of the **numerical variable** depend on the value of the **categorical variable**?

- Do the samples **differ** systematically in **mean** or **median** value?

There are MANY t-tests types, each one with their own equation: make sure you apply the one corresponding to the case you are studying

Two-sample t-test

Independent samples subjected to same measurement.

- **<u>Purpose</u>:** Test significance of **difference in means** of 2 samples of **numerical** variables. Samples refer to 2 different values of a **categorical** variable.

  e.g. Are men and women equally tall?

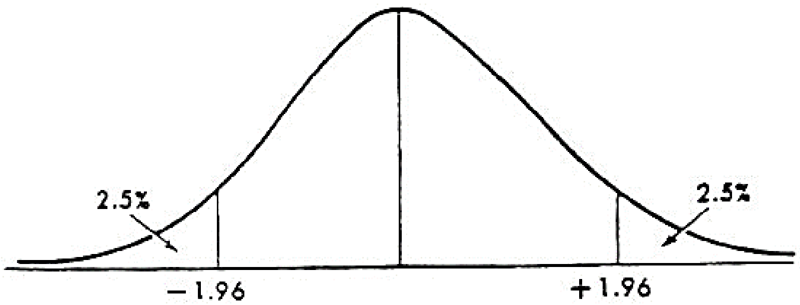
- **<u>Null hypothesis</u> (H_0):** There is no difference between the sample 1&2.

  There is **no** systematic **difference**. Sample means differ only by **chance**.

- **<u>Alternative hypothesis</u>** (**H_1**): The sample means are different.

  - Men and women are not equally tall
    (<u>but we're not saying who is taller</u>) (**2-tailed**)
  - Men are taller than women (**1-tailed**) OR
  - Women are taller than men (**1-tailed**)

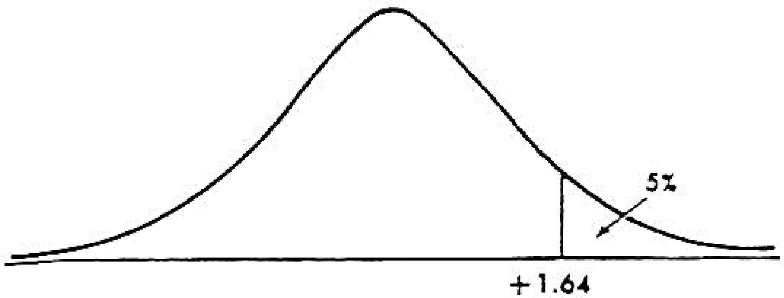
1 and 2 tailed tests



Two-tailed test at the 5 percent level.

2-tailed: difference can be in either direction (or distribution tail)

H_0: $\mu_1 = \mu_2$



One-tailed test at the 5 percent level.

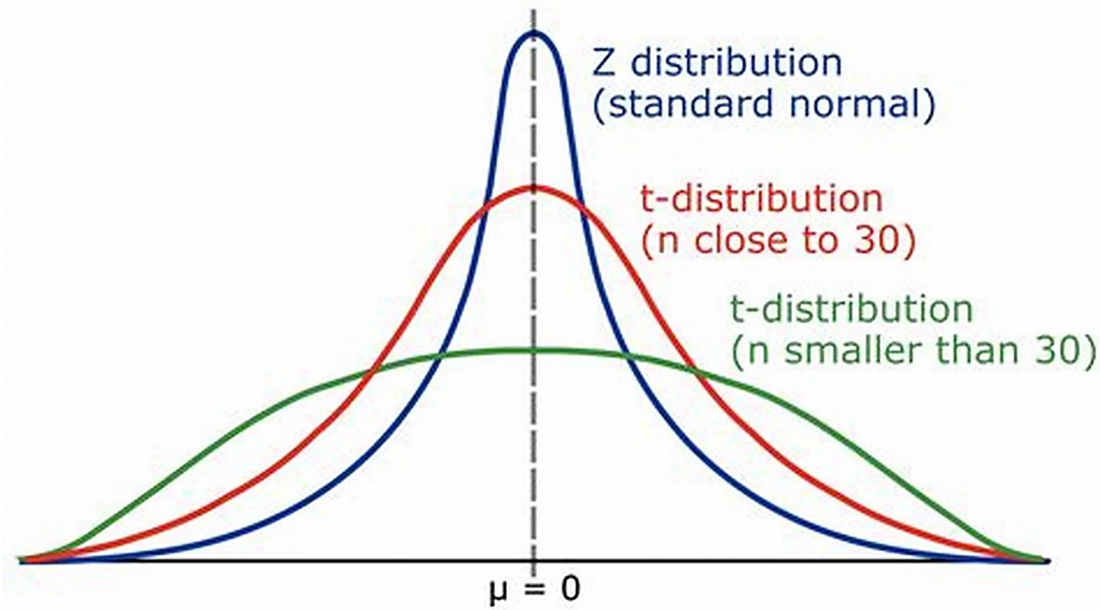
1-tailed: difference can only be in one direction (or distribution tail)

H_0: $\mu_1 > \mu_2$ or $\mu_1 < \mu_2$

Two-sample t-test

How 2-sample (independent) t-test works?

- Calculate test statistic d/SE$_{(d)}$ (**"sample t value"**) and compare with a critical t-value for the degree of freedom of the sample (related to number of measurements or data points)
- If $\mathbf{H_0}$ is true, this test statistic follows the t distribution (like a normal distribution but tails are slightly stretched out)



Formula for t statistic

Sample mean of sample 1

Sample mean of sample 2

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{(s_1^2/n_1) + (s_2^2/n_2)}}$$

Standard deviation of sample 1

Sample size of sample 1

Standard deviation of sample 2

Sample size of sample 2

Equation for t statistic: independent samples, equal variances

Critical Values of the Student's t Distribution

Taken from: https://www.itl.nist.gov/div898/handbook/eda/section3/eda3672.htm

The table in following pages contains critical values of the Student's t distribution.
The t distribution is symmetric so that $\mathbf{t_{1-\alpha,\nu} = -t_{\alpha,\nu}}$

The t table can be used for both one-tailed (also called one-sided), either lower or upper, and two-tailed (also two-sided) tests using the appropriate value of α.

The difference lies in finding out whether you are conducting a one-tailed test, or two-tailed test. For a one-tailed test, use the 1-α column (for example "0.95" for 95% confidence, α=0.05); for a two-tailed test, use the 1-α/2 column (for example "0.975" for 95% confidence, α=0.05) vs **the degrees of freedom for equal variance (= number of events-2)** for the t-critical values.

Decide whether you will be conducting a one-tailed or two-tailed test.
Calculate the t statistic for your samples. Find the critical t value and compare.
If t > critical t, then you reject the null hypothesis H_0, gaining confidence in your alternative hypothesis.

Question: what's the relevance of negative and positive values of t?

Critical values of Student's t distribution with v degrees of freedom

Probability less than the critical value ($t_{1-\alpha,v}$)

v	0.90	0.95	0.975	0.99	0.995	0.999
1.	3.078	6.314	12.706	31.821	63.657	318.313
2.	1.886	2.920	4.303	6.965	9.925	22.327
3.	1.638	2.353	3.182	4.541	5.841	10.215
4.	1.533	2.132	2.776	3.747	4.604	7.173
5.	1.476	2.015	2.571	3.365	4.032	5.893
6.	1.440	1.943	2.447	3.143	3.707	5.208
7.	1.415	1.895	2.365	2.998	3.499	4.782
8.	1.397	1.860	2.306	2.896	3.355	4.499
9.	1.383	1.833	2.262	2.821	3.250	4.296
10.	1.372	1.812	2.228	2.764	3.169	4.143
11.	1.363	1.796	2.201	2.718	3.106	4.024
12.	1.356	1.782	2.179	2.681	3.055	3.929
13.	1.350	1.771	2.160	2.650	3.012	3.852
14.	1.345	1.761	2.145	2.624	2.977	3.787
15.	1.341	1.753	2.131	2.602	2.947	3.733
16.	1.337	1.746	2.120	2.583	2.921	3.686
17.	1.333	1.740	2.110	2.567	2.898	3.646
18.	1.330	1.734	2.101	2.552	2.878	3.610
19.	1.328	1.729	2.093	2.539	2.861	3.579
20.	1.325	1.725	2.086	2.528	2.845	3.552
21.	1.323	1.721	2.080	2.518	2.831	3.527
22.	1.321	1.717	2.074	2.508	2.819	3.505
23.	1.319	1.714	2.069	2.500	2.807	3.485

$$\alpha = 0.05$$
$$\textit{Degrees of freedom } 9$$
$$1 - \textit{tailed } t_{critical} = 1.833$$

Critical values of Student's *t* distribution with *v* degrees of freedom

Probability less than the critical value ($t_{1-\alpha,\nu}$)

v	0.90	0.95	0.975	0.99	0.995	0.999
24.	1.318	1.711	2.064	2.492	2.797	3.467
25.	1.316	1.708	2.060	2.485	2.787	3.450
26.	1.315	1.706	2.056	2.479	2.779	3.435
27.	1.314	1.703	2.052	2.473	2.771	3.421
28.	1.313	1.701	2.048	2.467	2.763	3.408
29.	1.311	1.699	2.045	2.462	2.756	3.396
30.	1.310	1.697	2.042	2.457	2.750	3.385
31.	1.309	1.696	2.040	2.453	2.744	3.375
32.	1.309	1.694	2.037	2.449	2.738	3.365
33.	1.308	1.692	2.035	2.445	2.733	3.356
34.	1.307	1.691	2.032	2.441	2.728	3.348
35.	1.306	1.690	2.030	2.438	2.724	3.340
36.	1.306	1.688	2.028	2.434	2.719	3.333
37.	1.305	1.687	2.026	2.431	2.715	3.326
38.	1.304	1.686	2.024	2.429	2.712	3.319
39.	1.304	1.685	2.023	2.426	2.708	3.313
40.	1.303	1.684	2.021	2.423	2.704	3.307
41.	1.303	1.683	2.020	2.421	2.701	3.301
42.	1.302	1.682	2.018	2.418	2.698	3.296
43.	1.302	1.681	2.017	2.416	2.695	3.291
44.	1.301	1.680	2.015	2.414	2.692	3.286
45.	1.301	1.679	2.014	2.412	2.690	3.281
46.	1.300	1.679	2.013	2.410	2.687	3.277
47.	1.300	1.678	2.012	2.408	2.685	3.273

v	0.90	0.95	0.975	0.99	0.995	0.999
48.	1.299	1.677	2.011	2.407	2.682	3.269
49.	1.299	1.677	2.010	2.405	2.680	3.265
50.	1.299	1.676	2.009	2.403	2.678	3.261
51.	1.298	1.675	2.008	2.402	2.676	3.258
52.	1.298	1.675	2.007	2.400	2.674	3.255
53.	1.298	1.674	2.006	2.399	2.672	3.251
54.	1.297	1.674	2.005	2.397	2.670	3.248
55.	1.297	1.673	2.004	2.396	2.668	3.245
56.	1.297	1.673	2.003	2.395	2.667	3.242
57.	1.297	1.672	2.002	2.394	2.665	3.239
58.	1.296	1.672	2.002	2.392	2.663	3.237
59.	1.296	1.671	2.001	2.391	2.662	3.234
60.	1.296	1.671	2.000	2.390	2.660	3.232
61.	1.296	1.670	2.000	2.389	2.659	3.229
62.	1.295	1.670	1.999	2.388	2.657	3.227
63.	1.295	1.669	1.998	2.387	2.656	3.225
64.	1.295	1.669	1.998	2.386	2.655	3.223
65.	1.295	1.669	1.997	2.385	2.654	3.220
66.	1.295	1.668	1.997	2.384	2.652	3.218
67.	1.294	1.668	1.996	2.383	2.651	3.216
68.	1.294	1.668	1.995	2.382	2.650	3.214
69.	1.294	1.667	1.995	2.382	2.649	3.213
70.	1.294	1.667	1.994	2.381	2.648	3.211
71.	1.294	1.667	1.994	2.380	2.647	3.209
72.	1.293	1.666	1.993	2.379	2.646	3.207
73.	1.293	1.666	1.993	2.379	2.645	3.206
74.	1.293	1.666	1.993	2.378	2.644	3.204
75.	1.293	1.665	1.992	2.377	2.643	3.202
76.	1.293	1.665	1.992	2.376	2.642	3.201

v	0.90	0.95	0.975	0.99	0.995	0.999
77.	1.293	1.665	1.991	2.376	2.641	3.199
78.	1.292	1.665	1.991	2.375	2.640	3.198
79.	1.292	1.664	1.990	2.374	2.640	3.197
80.	1.292	1.664	1.990	2.374	2.639	3.195
81.	1.292	1.664	1.990	2.373	2.638	3.194
82.	1.292	1.664	1.989	2.373	2.637	3.193
83.	1.292	1.663	1.989	2.372	2.636	3.191
84.	1.292	1.663	1.989	2.372	2.636	3.190
85.	1.292	1.663	1.988	2.371	2.635	3.189
86.	1.291	1.663	1.988	2.370	2.634	3.188
87.	1.291	1.663	1.988	2.370	2.634	3.187
88.	1.291	1.662	1.987	2.369	2.633	3.185
89.	1.291	1.662	1.987	2.369	2.632	3.184
90.	1.291	1.662	1.987	2.368	2.632	3.183
91.	1.291	1.662	1.986	2.368	2.631	3.182
92.	1.291	1.662	1.986	2.368	2.630	3.181
93.	1.291	1.661	1.986	2.367	2.630	3.180
94.	1.291	1.661	1.986	2.367	2.629	3.179
95.	1.291	1.661	1.985	2.366	2.629	3.178
96.	1.290	1.661	1.985	2.366	2.628	3.177
97.	1.290	1.661	1.985	2.365	2.627	3.176
98.	1.290	1.661	1.984	2.365	2.627	3.175
99.	1.290	1.660	1.984	2.365	2.626	3.175
100.	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.090

How 2-sample t-test works

- Compare test statistic with tabulated **critical value of *t*** for chosen **significance level α**, e.g. α=0.05, z=1.96 (95% confidence):

$$\boldsymbol{Reject\ H_0\ if\ ovserved\ t > t_{critical}, p < 0.05,}$$
$$\boldsymbol{otherwise\ we\ can't\ reject\ H_0.}$$

- Or (in Excel other software) the computer uses *t* tables to compute p-value (probability of getting observed difference between samples by **chance** if $\mathbf{H_0}$ were true), as well, as getting t and critical t for the given degrees of freedom:

$$\boldsymbol{Reject\ H_0\ if\ calculated\ p\ value\ < 0.05, otherwise\ we\ can't\ reject\ H_0.}$$

- Report the outcome in scientific terms:

e.g. "the mean of men's height was statistically significant from the mean of women's height **(2-sample t-test, $p < 0.05$)**."

T-test conditions

t-test **assumes** that the sample means follows a **normal distribution** when $\mathbf{H_0}$ is true:

- Valid if **population is normal distribution** (in which case samples will appear approximately normal)
- Also valid for **non-normal** populations (and samples) **if** sample size is **large** (> 30 per sample)
- Should **not** be used if samples are both small & non-normal.
- Unreliable if samples are too small to tell whether normally distributed.

Summary

z-test - sample different from a population

(are the new results different from the 'norm'?)

t-test - samples different from each other

(are two samples different from the each other?)

EXP2-1 Schedule – P Groups

Date:	Time:	Place:	Activity:
Monday 20th	14:00-14:45	QMES Floor 1	Introduction to QXU5017 Introduction to EXP2-1 Part1
Tuesday 21st	10:30-12:10		Group Meeting 1 (Correct data presenting)
Friday 24th	14:00-15:40		Group Meeting 2 (Write Report)
Monday 27th	14:00-15:40	QMES Floor 1	Introduction to EXP2-1 Part2
Tuesday 28th	10:30-12:10		Group Meeting 3 (Different sample comparison)
Friday 3rd	14:00-15:40		Group Meeting 4 (Complete report)
Sunday 5th	17:00	QMPlus	Report submission

EXP2-1 Schedule – M Groups

Date:	Time:	Place:	Activity:
Monday 20th	10:30-11:15	QMES Floor 1	Introduction to QXU5017 Introduction to EXP2-1 Part1
Tuesday 21st	14:00-15:40		Group Meeting 1 (Correct data presenting)
Friday 24th	10:30-12:10		Group Meeting 2 (Write Report)
Monday 27th	10:30-12:10	QMES Floor 1	Introduction to EXP2-1 Part2
Tuesday 28th	14:00-15:40		Group Meeting 3 (Different sample comparison)
Friday 3rd	10:30-12:10		Group Meeting 4 (Complete report)
Sunday 5th	17:00	QMPlus	Report submission